# Structure-Exploiting Factor-Risk Quadratic Programming for One-Shot and Ladder Portfolio Construction

QUARCC
Quantitative Research Note
November 2026

**Abstract.** Equity long–short portfolio construction repeatedly solves convex quadratic programs whose covariance decomposes into a low-rank factor component and a diagonal specific component, $\boldsymbol{\Sigma} = \mathbf{B}\mathbf{F}\mathbf{B}^\top + \operatorname{diag}(\mathbf{d})$. Modelled through general-purpose expression graphs and solved as generic conic programs, the problem spends much of its runtime assembling and canonicalising. We describe a primal–dual Mehrotra predictor–corrector interior-point method that keeps factor risk factored through the Newton step and reuses static constraint data across rebalance ladders. Slack elimination condenses the Karush–Kuhn–Tucker system into a diagonal-plus-low-rank matrix of rank $r = k + m$, with $k$ risk factors and $m$ two-sided exposure bands. The Woodbury identity and a small Schur complement reduce per-iteration linear algebra from $O(n^3)$ to $O(nr^2 + r^3)$. On a synthetic 1,000-security book with $r = 43$, single-thread solve time falls from 1,275.5 ms under CVXPY with Clarabel and 97.7 ms under a hand-built sparse Clarabel model to 25.3 ms, with an objective gap of $5.3 \times 10^{-6}$. Across a 36-rung ladder at $n = 500$ the method is $18.9\times$ faster than rebuilding through CVXPY and $3.1\times$ faster than the sparse model. In a synthetic 12-month run of a four-stage market-neutral construction sequence, end-to-end runtime falls from 8.73 s to 2.93 s. The speedup comes with small but nonzero differences in weights, which we report.

## Problem Formulation

Cross-sectional equity strategies trade off expected return against factor and specific variance, subject to limits on positions, exposures, leverage, and turnover. A research process then solves the same problem many times: across gross targets, support sizes, and formation dates. The numerical kernel is a convex QP in weights $\mathbf{x} \in \mathbb{R}^n$:

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2}\mathbf{x}^\top \mathbf{P}\mathbf{x} + \mathbf{q}^\top \mathbf{x} + c_0 \\ \text{s.t.} \quad & \boldsymbol{\ell} \le \mathbf{x} \le \mathbf{u}, \\ & \mathbf{g}_\ell \le \mathbf{G}\mathbf{x} \le \mathbf{g}_u, \quad \mathbf{A}\mathbf{x} = \mathbf{b}. \end{aligned} \tag{1}$$

Box bounds encode position caps, liquidity limits, and fixed signs. $\mathbf{G}$ holds exposure bands, and $\mathbf{A}$ holds equalities such as gross leverage for a fixed sign vector. For exposures $\mathbf{B} \in \mathbb{R}^{n\times k}$, factor covariance $\mathbf{F}$, and specific variances $\mathbf{d}$, the covariance is

$$\boldsymbol{\Sigma} = \mathbf{B}\mathbf{F}\mathbf{B}^\top + \operatorname{diag}(\mathbf{d}). \tag{2}$$

With risk aversion $\gamma$ and quadratic trading-cost weights $\boldsymbol{\tau}_2$, the operator keeps a factored form,

$$\mathbf{P} = \mathbf{U}\operatorname{diag}(\mathbf{c})\,\mathbf{U}^\top + \operatorname{diag}(\mathbf{p}_d), \tag{3}$$

where $\mathbf{F} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^\top$, $\mathbf{U} = \mathbf{B}\mathbf{V}\boldsymbol{\Lambda}^{1/2}$, $\mathbf{c} = \gamma \mathbf{1}_k$, and $\mathbf{p}_d = \gamma\mathbf{d} + \boldsymbol{\tau}_2$.

Through CVXPY (Diamond and Boyd, 2016) with Clarabel (Goulart and Chen, 2024), two overheads recur on every solve: forming dense $\boldsymbol{\Sigma}$ costs $O(n^2 k)$, and canonicalising the dense quadratic form leads to an $O(n)$-dimensional factorisation per iteration. A hand-built sparse model with auxiliary factor variables $\mathbf{f} = \mathbf{U}^\top\mathbf{x}$ avoids most of this, and it is the fairer baseline, but it repeats cone setup and general-purpose factorisation on each call.

Table 1. Single-thread benchmarks on a 2-vCPU Linux VM (tol $= 10^{-8}$, medians of repeated runs, BLAS pinned to one thread, one core). Problems are synthetic. The baseline is CVXPY with Clarabel or, for the 100-name book, SLSQP, and accuracy is measured against it; "sparse" is the hand-built sparse Clarabel model; dashes denote quantities not measured.

| Workload | Solver path | Median (ms) | vs. baseline | vs. sparse | Objective gap | Max $\lvert\Delta x_i\rvert$ |
|---|---|---|---|---|---|---|
| One-shot ($n = 300$, $r = 43$) | CVXPY + Clarabel | 88.3 | 1.0× | — | 0 | 0 |
| | Sparse Clarabel | 26.4 | 3.3× | 1.0× | 2.1e-9 | 1.2e-5 |
| | This method (12 iterations) | **10.0** | **8.8**× | **2.6**× | 2.1e-6 | 4.4e-4 |
| One-shot ($n = 1000$, $r = 43$) | CVXPY + Clarabel | 1,275.5 | 1.0× | — | 0 | 0 |
| | Sparse Clarabel | 97.7 | 13.1× | 1.0× | 2.1e-10 | 2.3e-7 |
| | This method (18 iterations) | **25.3** | **50.4**× | **3.9**× | 5.3e-6 | 5.3e-4 |
| Ladder ($n = 500$, 36 solves, per solve) | CVXPY rebuild | 285.5 | 1.0× | — | 0 | 0 |
| | Sparse Clarabel | 47.2 | 6.0× | 1.0× | — | — |
| | This method, compiled | **15.1** | **18.9**× | **3.1**× | 3.2e-6 | 5.5e-4 |
| Market-neutral book ($n = 200$) | CVXPY + Clarabel | 46.3 | 1.0× | — | 0 | 0 |
| | This method | **9.8** | **4.7**× | — | 5.3e-7 | 4.5e-4 |
| Signed book ($n = 100$, dense risk) | SciPy SLSQP | 150.6 | 1.0× | — | 0 | 0 |
| | This method | **38.6** | **3.9**× | — | — | 7.4e-5 |

## Factored Interior-Point Kernel

**Rank-one augmentations.** Penalties on linear aggregates preserve (3). A dollar-neutrality penalty $\frac{1}{2}\rho_{\text{net}}(\mathbf{1}^\top\mathbf{x})^2$ appends the column $\mathbf{1}$ to $\mathbf{U}$ and the weight $\rho_{\text{net}}$ to $\mathbf{c}$. A target penalty $\frac{1}{2}\lambda_j(\mathbf{v}_j^\top\mathbf{x} - \beta_{0,j})^2$ appends $\mathbf{v}_j$ and $\lambda_j$, shifts $\mathbf{q}$ by $-\lambda_j\beta_{0,j}\mathbf{v}_j$, and adds $\frac{1}{2}\lambda_j\beta_{0,j}^2$ to $c_0$.

**Primal–dual system.** We introduce positive slacks $\mathbf{s}_{\text{hi}}, \mathbf{s}_{\text{lo}}, \mathbf{s}_{gu}, \mathbf{s}_{gl}$, multipliers $\mathbf{z}_{\text{hi}}, \mathbf{z}_{\text{lo}}, \mathbf{z}_{gu}, \mathbf{z}_{gl}$, and equality multipliers $\mathbf{y}$. Infinite bounds are masked out. The residuals are

$$\begin{aligned} \mathbf{r}_d &= \mathbf{P}\mathbf{x} + \mathbf{q} + (\mathbf{z}_{\text{hi}} - \mathbf{z}_{\text{lo}}) + \mathbf{G}^\top(\mathbf{z}_{gu} - \mathbf{z}_{gl}) + \mathbf{A}^\top\mathbf{y}, \\ \mathbf{r}_{p,\text{hi}} &= \mathbf{x} + \mathbf{s}_{\text{hi}} - \mathbf{u}, \qquad \mathbf{r}_{p,\text{lo}} = -\mathbf{x} + \mathbf{s}_{\text{lo}} + \boldsymbol{\ell}, \\ \mathbf{r}_{p,gu} &= \mathbf{G}\mathbf{x} + \mathbf{s}_{gu} - \mathbf{g}_u, \quad \mathbf{r}_{p,gl} = -\mathbf{G}\mathbf{x} + \mathbf{s}_{gl} + \mathbf{g}_\ell, \\ \mathbf{r}_e &= \mathbf{A}\mathbf{x} - \mathbf{b}, \end{aligned} \tag{4}$$

where $\mathbf{P}\mathbf{x} = \mathbf{U}(\mathbf{c} \odot (\mathbf{U}^\top\mathbf{x})) + \mathbf{p}_d \odot \mathbf{x}$ costs $O(nk)$.

**Woodbury normal equations.** With barrier weights $\mathbf{w}_\bullet = \mathbf{z}_\bullet \oslash \mathbf{s}_\bullet$, pivoting out $(\Delta\mathbf{s}, \Delta\mathbf{z})$ reduces each Newton step to

$$\mathbf{M}\,\Delta\mathbf{x} + \mathbf{A}^\top\Delta\mathbf{y} = \mathbf{r}_{\text{rhs}}, \qquad \mathbf{A}\,\Delta\mathbf{x} = -\mathbf{r}_e, \tag{5}$$

where

$$\begin{aligned} \mathbf{M} &= \mathbf{P} + \operatorname{diag}(\mathbf{w}_{\text{hi}} + \mathbf{w}_{\text{lo}}) + \mathbf{G}^\top\operatorname{diag}(\mathbf{w}_{gu} + \mathbf{w}_{gl})\mathbf{G} \\ &= \mathbf{H} + \mathbf{U}_{\text{aug}}\operatorname{diag}(\mathbf{c}_{\text{aug}})\,\mathbf{U}_{\text{aug}}^\top. \end{aligned} \tag{6}$$

Here $\mathbf{H} = \operatorname{diag}(\mathbf{p}_d + \mathbf{w}_{\text{hi}} + \mathbf{w}_{\text{lo}} + \varepsilon\mathbf{1})$ with $\varepsilon = 10^{-12}$, the matrix $\mathbf{U}_{\text{aug}} = [\mathbf{U}, \mathbf{G}^\top]$ is fixed before the iteration loop, and $\mathbf{c}_{\text{aug}} = \max([\mathbf{c};\, \mathbf{w}_{gu} + \mathbf{w}_{gl}], \varepsilon)$. By the Sherman–Morrison–Woodbury identity (Woodbury, 1950), applying $\mathbf{M}^{-1}$ needs only an $r \times r$ capacitance system:

$$\mathbf{K} = \operatorname{diag}(\mathbf{c}_{\text{aug}})^{-1} + \mathbf{U}_{\text{aug}}^\top\mathbf{H}^{-1}\mathbf{U}_{\text{aug}}, \tag{7}$$

$$\mathbf{M}^{-1}\mathbf{V} = \mathbf{H}^{-1}\mathbf{V} - \mathbf{H}^{-1}\mathbf{U}_{\text{aug}}\mathbf{K}^{-1}\mathbf{U}_{\text{aug}}^\top\mathbf{H}^{-1}\mathbf{V}. \tag{8}$$

Assembling $\mathbf{K}$ costs $O(nr^2)$ and factorising it costs $O(r^3)$. With $p$ equalities, the $p \times p$ Schur complement $\mathbf{S} = \mathbf{A}\mathbf{M}^{-1}\mathbf{A}^\top$ gives

$$\begin{aligned} \Delta\mathbf{y} &= \mathbf{S}^{-1}\left(\mathbf{A}\mathbf{M}^{-1}\mathbf{r}_{\text{rhs}} + \mathbf{r}_e\right), \\ \Delta\mathbf{x} &= \mathbf{M}^{-1}\mathbf{r}_{\text{rhs}} - (\mathbf{M}^{-1}\mathbf{A}^\top)\Delta\mathbf{y}. \end{aligned} \tag{9}$$

**Predictor–corrector step.** Following Mehrotra (1992), each iteration computes an affine-scaling direction, the maximum step that keeps slacks and multipliers at no less than 1% of their current values, and the centering parameter $\sigma = \min(\max(\mu_{\text{aff}}/\mu, 0), 1)^3$. The combined step replaces the complementarity term with $\mathbf{s} \odot \mathbf{z} + \Delta\mathbf{s}_{\text{aff}} \odot \Delta\mathbf{z}_{\text{aff}} - \sigma\mu\mathbf{1}$ and reuses $\mathbf{K}$, $\mathbf{M}^{-1}\mathbf{A}^\top$, and $\mathbf{S}$. Convergence is declared when all residual norms and $\mu$ are below tol, by default $10^{-8}$.

## Outer Loops and Ladder Compilation

**Ladder compilation.** Across a ladder, $\mathbf{U}$, $\mathbf{G}$, $\mathbf{A}$, $\mathbf{U}_{\text{aug}}$, and the active-bound masks are invariant, while $\mathbf{q}$ and $\mathbf{b}$ vary. We cache the static data once, and each later rung enters the interior-point loop directly.

**Ex-ante risk cap.** For a hard budget $\mathbf{x}^\top\boldsymbol{\Sigma}\mathbf{x} \le \sigma_{\max}^2$, we solve at the baseline $\gamma$. If the budget is exceeded, we bracket the multiplier shift $\lambda$ by doubling and bisect on $\gamma + \lambda$. Each inner solve warm-starts the primal vector and the equality dual, and resets barrier slacks and inequality duals to avoid numerical blow-up on constraints that were nearly active.

**L1 turnover.** A penalty $\sum_i \tau_{1,i}|x_i - x_{i,\text{prev}}|$ is handled without $2n$ splitting variables by iteratively reweighted least squares. With previous trades $\mathbf{q}^{(t-1)}$ the smoothed majoriser weights are

$$a_i^{(t)} = \frac{\tau_{1,i}}{\sqrt{\left(q_i^{(t-1)}\right)^2 + \epsilon^2}}, \qquad \epsilon = 10^{-8}, \tag{10}$$

which are added to the diagonal. Each pass remains a diagonal-plus-low-rank QP of unchanged dimension and rank.

**Soft bands.** One-sided soft bands default to a quadratic slack penalty with $\rho_{\text{soft}} = 10^4$; exact linear slacks are available at the cost of more iterations on degenerate books.

## Fast Screening Estimators

**Loose-tolerance preview.** Stopping at tol $= 10^{-3}$ and six iterations gives an approximate solution for screening and warm starts (Table 2, a separate timing experiment). The objective gap is 2.8–3.3 $\times\, 10^{-3}$ and the maximum weight difference is at most $1.9 \times 10^{-2}$, so a preview ranks candidates but is not a book. Allowing up to 16 iterations tightens these to $7.5 \times 10^{-5}$ and $4 \times 10^{-3}$.

Table 2. Exact solve (tol $= 10^{-8}$) vs. six-iteration preview (tol $= 10^{-3}$), in milliseconds.

| $n$ | Exact | Preview | Speedup | Obj. gap | Max $\lvert\Delta x\rvert$ |
|---|---|---|---|---|---|
| 300 | 10.0 | 3.7 | 2.7× | $3.3 \times 10^{-3}$ | $1.6 \times 10^{-2}$ |
| 1000 | 24.7 | 4.9 | 5.0× | $2.8 \times 10^{-3}$ | $1.9 \times 10^{-2}$ |

**Constraint-contact screen.** Motivated by jamming in constrained perceptrons (Franz and Parisi, 2016), we define the contact ratio of a sign-fixed candidate book:

$$\rho_{\text{contact}} = \frac{\#\{i : x_i \text{ at a bound}\} + \#\{\text{active bands}\}}{n_{\text{selected}}}. \tag{11}$$

On 16 labelled formations of a market-neutral construction problem, all successful first rungs of a gross ladder had $\rho_{\text{contact}} < 0.47$, while the two failures had 0.59 and 0.65. A two-parameter logistic with threshold 0.53 and slope 100 turns this into a ranking score. The two failures come from a single calendar month, so this is a screening heuristic and not a probability.

## Empirical Scaling

Table 1 separates two effects at $n = 1{,}000$. Removing the expression graph and dense canonicalisation gives 13.1× (1,275.5 ms to 97.7 ms). Replacing general conic factorisation with the $43 \times 43$ Woodbury system gives a further 3.9× (to 25.3 ms), for 50.4× in total. With rank fixed, larger signed stage-0 books with no minimum position solve in about 80 ms at $n = 2{,}000$ and about 200 ms at $n = 5{,}000$ .

In a synthetic 12-month run of a four-stage market-neutral construction sequence ($n = 400$, 10 sectors, 5 styles; Table 3), a single base QP falls from 86.8 ms to 20.1 ms ( 4.3×, maximum weight difference $3.5 \times 10^{-7}$). Supported problems use the solver, and unsupported features fall back to Clarabel. The last two rows of Table 3 also substitute approximations: a fixed-sign preview in the first pass and symmetric L1 trade costs in the last stage. They trade exactness for speed, with maximum weight differences near $3 \times 10^{-3}$ and $2 \times 10^{-3}$.

Table 3. Four-stage construction runtime in seconds, 12 synthetic formations ($n = 400$).

| Solver path | Stage 1 | Stage 2 | Stage 3 | Stage 4 | Total |
|---|---|---|---|---|---|
| Clarabel baseline | 0.690 | 2.572 | 2.652 | 2.812 | 8.726 |
| This method, final QPs | 0.160 | 1.714 | 1.779 | 2.849 | 6.502 |
| + fixed-sign preview | 0.157 | 0.706 | 0.694 | 2.883 | 4.439 |
| + symmetric L1 costs | **0.160** | **0.673** | **0.698** | **1.399** | **2.930** |

## Limitations

First, all benchmarks use synthetic problems on one machine, so absolute times will differ on real data and other hardware. Second, the results measure computational performance and say nothing about investment outcomes. Third, the method assumes sign-fixed weights, so free-sign support selection with minimum positions still needs an outer ladder or a first-stage sign assignment. Fourth, L1 turnover reaches the exact objective to roughly $10^{-5}$, and the default quadratic soft-band penalty explains weight differences near $5 \times 10^{-4}$. Fifth, the contact screen rests on two failures from one month.

## Code Availability

The solver code is not publicly available.

## References


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